

FreshRetailNet-50K

A Benchmark for Censored Demand Estimation

BTP Project Team

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FreshRetailNet-50K is the first large-scale benchmark dataset designed for **censored demand estimation** in the fresh retail domain.

The Problem It Solves:

- In retail, when a product goes out of stock (stockout), you can no longer measure its true demand.
- This dataset provides the necessary ground-truth and context to predict actual demand.

Key Highlights:

- Sourced by Dingdong Inc., featuring 20% organically occurring stockout data.
- Deals with the unique challenges of perishable goods with short shelf lives.

Dataset Scale & Features

Scale & Scope:

- **Timeframe**: 90-day time series of detailed hourly sales data.
- **Volume**: 50,000 unique store-product pairs.
- **Coverage**: 898 stores across 18 major cities.
- **Products**: 865 diverse perishable SKUs.

Rich Feature Set: Highly granular hourly data including:

- **Sales & Stock**: Sale amounts, hourly sales, and stock status.
- **Events & Pricing**: Holiday flags, activity flags, and discounts.
- **Weather Data**: Precipitation, temperature, humidity, and wind levels.

Data Selection & Splitting

Unit of Analysis & Target Selection:

- Predict at the level of `city_id + store_id + product_id`.
- Top 15 SKUs selected using a weighted multi-factor statistical score:

$$0.40 \times \text{Sales} + 0.25 \times \text{Stockout Hours} + 0.20 \times \text{Stockout Days} + 0.15 \times \text{Sales Std}$$

Train/Validation Split:

- **Chronological**: Strict temporal split to prevent data leakage.
- **Training Set**: Historical data up to $T_{\max} - 15$ days (2.5 months).
- **Validation Set**: Final 15 days ($T_{\max} - 15 < t \leq T_{\max}$).
- **Standardization**: StandardScaler fitted strictly on the training set.

Modeling Framework & Evaluation

Threshold & Metrics:

- **Dynamic Thresholding:** We scan thresholds $\tau \in [0.10, 0.90]$ to find the optimal boundary τ^* that maximizes performance on the validation split.
- **Primary Metric: Hour-Level F1-Score.** Raw accuracy is misleading because operational hours are heavily dominated by in-stock status (up to 80:20). F1 ensures true stockouts are accurately captured.